

10 EVERYTHING MUST GO SOMEWHERE

I still remember her so vividly. "I'm so happy now that you are here again," she would say, smiling, clasping my hand in hers and trying unsuccessfully to sit up in her hospital bed. "I am so blessed to have you and Iris. And your sister of course. Did you know that woman at the entrance has nobody at all? I really feel for her."

And I would shake my head and smile at how she always put other people first. At how welcome she made me feel even though *she* was in the hospital. But then I would look at her, and my heart would break again, as I saw my mother lying there, fighting for air even though she had a tube with oxygen going up her nose. Overweight, high blood pressure, brittle bones. The increased infection risk that would do her in in the end. All from the prednisone they were forced to give her because of her failing lungs.

Did you know that 7 *million* people die, every year, and *hundreds of millions* suffer, from the particulate matter we put into the air by burning fuel? ([WHO, 2022](#)) Why is everybody talking about global warming, but so few people about the people who are suffering *right here, right now* from the same cause. I can't determine to what extent my mother's death and suffering was caused by burning fossil fuel. But emotionally, it affects me even stronger than the obligation that I feel to future generations.

I didn't plan to write that. But sometimes a number like 7 million only becomes real through one.

We have been talking about the raw materials we need as the inflow of the bathtub. We have been talking about the bathtub itself when we considered its land use. Now it's time to look at all the unwanted stuff that flows out of the bathtub as we produce the energy that our civilization depends upon.

10.1 Barry Commoner's Ecological Laws

The title of this chapter honours Barry Commoner, an American biologist who helped launch the modern environmental movement. ([Markowitz, 2018](#)) By all accounts he was as kind as he was brilliant and he inspired countless students, often to the point of giving him a standing ovation after a lecture. In 1971, he wrote a book called *The Closing Circle* in which he posits four "laws of ecology." ([Markowitz, 2018](#)) The first was: *everything is connected to everything else*. Commoner understood systems thinking. He saw the feedback loops, the stocks and flows, the way a change in one part of a system ripples through every other part. He would have approved of our bathtub metaphor.

His second law gives this chapter its title: *everything must go somewhere*. Matter doesn't vanish. When you burn coal, the carbon becomes CO₂ in the atmosphere. When you spread fertilizer, excess nitrogen runs off into rivers. When smoke rises from a chimney, it enters someone's lungs. There is no "away."

His third law, *nature knows best*, is often misunderstood. Commoner was not a primitivist rejecting all technology. He opposed the use of synthetic chemicals that ecosystems cannot break down: DDT accumulating in eagle eggs, PCBs persisting in rivers, plastics that nothing can digest. Materials that evolution never prepared nature to handle.

His fourth law, *there is no such thing as a free lunch*, warned against the fossil fuel economy specifically: the idea that we could draw down ancient stocks without consequence. Commoner saw the solution clearly. In 1976, he wrote *The Poverty of Power*, arguing for solar energy as the alternative to both nuclear and fossil fuels. (Commoner, 1976) He understood that sunlight is a flow, not a stock. That capturing it doesn't deplete anything. That a solar economy could honour all four of his laws. He was fifty years ahead of his time.

What Commoner couldn't foresee was the learning curve. He advocated for solar when solar panels cost five hundred times more than what they cost today. (Ritchie, 2023) He was right about the destination but had no way to predict how fast we would get there. Now we have arrived. The technology he championed is finally cheap enough to win. Reading an interview with him from 2007, five years before his death, I hope and think he would have been delighted with this book. (AAAS, 2012)

The question in this chapter is whether we will be able to honour his second law: whether we will stop treating the atmosphere, the water, and the soil as bottomless sinks for our waste.

10.1.1 What We Breathe

Let me put some numbers around the seven million.

The World Health Organization and the Lancet have studied this for decades. Their conclusion: air pollution kills roughly seven to nine million people every year. (Ritchie, 2021) That makes it one of the leading causes of death globally, exceeding malaria, tuberculosis, and HIV/AIDS combined. (Landrigan, 2018) Ninety-nine percent of the world's population breathes air that exceeds WHO guideline limits. (WHO, 2022) Not people in particularly polluted cities. Nearly everyone, everywhere.

This is not a distant problem affecting faraway places. Air pollution shortens lives in Amsterdam and London and Los Angeles. It correlates with dementia, depression, and reduced cognitive development in children. (Peters, 2019) It is linked to heart disease, stroke, lung cancer, and of course the respiratory diseases that plagued my mother. (WHO, 2022)

Roughly half of these deaths, approximately 4.2 million, come from outdoor air pollution. (WHO, 2024) The primary culprit is particulate matter, especially the tiny particles called PM2.5. These are particles less than 2.5 micrometers in diameter, small enough to pass through your lungs into your bloodstream. (EPA, 2025) You cannot see them. You cannot smell them. But they accumulate in your body and cause damage.

Where does PM2.5 come from? Mostly from burning things. Coal power plants. Diesel trucks and buses. Industrial furnaces. The combustion that powers modern civilization leaves a residue in the air, and that residue enters our lungs.

Coal is the worst offender. A single large coal plant can emit thousands of tonnes of particulates, sulfur dioxide, and nitrogen oxides per year. (UCS, 2017) The areas downwind have elevated rates of asthma, bronchitis, heart attacks, and premature death. (EPA, 2022) In China, the coal boom of the early 2000s created air so thick in some cities that people wore masks long before Covid made it fashionable. India faces similar struggles. Even in Europe and North America, where coal is declining, the legacy plants that remain still kill. And of course the exhaust from road traffic is also a big contributor.

The other half of air pollution deaths, roughly 3.2 million, come from indoor air pollution. (WHO, 2024) Over two billion people still cook with solid fuels: wood, charcoal, dung, coal. (WHO, 2024) The smoke fills their homes. Women and children are most exposed because they spend the most time near the cooking fire. Natural gas is much cleaner but make no mistake, it's a silent killer too. It is estimated that it kills around 19 thousand people who cook with it per year in the US (Kashtan, 2024), and around 40 thousand in Europe. (Seals, 2024) If we apply the same ratio to all people cooking with gas worldwide, that would mean at least hundreds of thousands of people killed every year. A simple solar panel with a battery and an electric cooktop could eliminate this exposure entirely. The technology exists. The cost is falling. The only question is how many people will suffer and die before we deploy it.

10.1.2 The Climate Bathtub

You have probably heard the climate argument a thousand times, so let's keep it short and make it a bit different. The bathtub metaphor makes it simple. Carbon dioxide in the atmosphere is a stock. Emissions from burning fossil fuels contribute to the inflow. Natural processes, forests absorbing carbon, oceans dissolving it, are the outflow.

For roughly ten thousand years, inflow and outflow were balanced. The concentration of CO₂ in the atmosphere stayed around 280 parts per million. (NOAA, 2025) The climate stayed relatively stable. Human civilization developed in that stable window.

Then we started burning coal, oil, and gas in ever-increasing quantities. The inflow vastly exceeds the outflow. The CO₂ stock in the atmosphere keeps rising. We've now crossed 420 parts per million (PPM), higher than at any point in at least 800,000 years. (NOAA, 2025) We call the result "global warming."

But we are not done yet. If things go on as planned we will reach 540 PPM before we are done. (Meinshausen, 2020) That level was last reached 15 million years ago (Steinthorsdottir, 2021) when the earth was three to four degrees hotter (You, 2009) and the sea level up to 40 meters higher. (Levy, 2016)

However, if we play our cards right, and we stop emitting CO₂ soon, this short spike will be partly neutralized by natural processes absorbing the CO₂ again. After that we could also

build a kind of thermostat for the earth in the form of carbon capture devices that regulate the temperature by capturing excess CO₂. But that will already be very expensive. Doing it while we keep emitting CO₂ would be prohibitively expensive and unrealistic. So how do we stop emitting green house gasses?

Approximately 76 percent of greenhouse gas emissions come from energy use: electricity, transport, heating, industry. (WRI, 2024) This means that if you solve energy, you solve three-quarters of the climate problem. Not all of it. But most of it. The transition to solar and wind addresses this directly. Solar panels and wind turbines produce zero emissions while operating. There are emissions embedded in manufacturing, in mining the materials, in transporting and installing the equipment. But life-cycle analyses consistently show that solar produces 20 to 50 times less CO₂ per kilowatt-hour than coal. (Schlömer, 2014) The emissions required to build a solar panel are paid back in one to two years of operation. (Mirletz, 2024) Then it runs clean for another 28 years or more. (DOE, 2025) Once we start producing solar and wind with solar and wind, the CO₂ emissions become zero.

Build the system once. Run it clean for decades. Recycle the materials. Repeat.

That leaves the remaining 24 percent: agriculture, land use, and certain industrial processes. (WRI, 2024) Electrifying energy doesn't touch these directly. But they lead us somewhere unexpected.

10.1.3 The Invisible Flood

Fossil fuels dominate air pollution deaths and climate change. Once you have replaced fossil fuel with solar and wind, the largest remaining emitter (by a mile) will be agriculture. Partly from "land use change" (cutting down trees) and partly from belching ruminants. Yes. Really.

But there's another reason we must look at food. Commoner's laws remind us that *everything* must go somewhere. There are patterns of pollution that most people seldom think about, pollution that transgresses planetary boundaries even more severely than carbon dioxide. And the biggest driver of this pollution is not energy. It is food.

Consider nitrogen. Every living thing needs nitrogen to build proteins and DNA. The atmosphere is 78 percent nitrogen gas (NOAA, 2024), but most organisms cannot use it in that form. For most of history, usable nitrogen was scarce. Farmers relied on manure, crop rotation, and the slow work of nitrogen-fixing bacteria in legume roots.

Then in 1909, Fritz Haber discovered how to pull nitrogen from the air and combine it with hydrogen to make ammonia. Carl Bosch scaled up the process industrially. (Britannica, 2024) The Haber-Bosch process, as it came to be known, is one of the most consequential inventions in human history. It enabled the production of synthetic fertilizer at massive scale. Food production soared. The population explosion of the twentieth century would not have been possible without it. (Smil, 1999) Some historians estimate that half of the nitrogen atoms in your body came through a Haber-Bosch plant. (Smil, 2001)

So far, so miraculous. The problem is what happens to the nitrogen we spread.

Less than half of applied nitrogen fertilizer actually ends up in crops. (You, 2023) The rest leaches into groundwater, runs off into rivers, and eventually reaches the ocean. There it triggers algal blooms: explosions of plant growth that consume all the oxygen in the water when they decompose. Marine life suffocates. The result is a "dead zone," an area where almost nothing can live.

The Gulf of Mexico has a dead zone the size of New Jersey that appears every summer, fed by nitrogen washing down the Mississippi from Midwestern farms. (NOAA, 2024) The Baltic Sea has similar problems. So do estuaries and coastal waters around the world.

Some of the excess nitrogen also converts to nitrous oxide, a greenhouse gas roughly 300 times more potent than CO₂ per molecule. (Tian, 2024) Agriculture, primarily through nitrogen fertilizer, is the main source of nitrous oxide emissions. (Tian, 2024)

Phosphorus tells a parallel story. It is essential for life, with no substitute. (Cordell, 2011) We mine it from ancient deposits and spread it on fields. Much of it washes into waterways, causing the same eutrophication problems as nitrogen. Unlike nitrogen, phosphorus cannot be synthesized from the air. We are drawing down a finite stock. Some analysts worry about "peak phosphorus" in the coming decades. (Cordell, 2011)

Then there are the chemicals that Commoner warned about. Pesticides designed to kill insects don't stay on the fields where they're sprayed. They spread through ecosystems. Studies have documented dramatic declines in flying insect biomass in some regions, sometimes over 75 percent in a few decades. (Hallmann, 2017) The causes are complex, but pesticide exposure is one of them. Neonicotinoid insecticides have been linked to declining bee populations. (Woodcock, 2016) We are loading ecosystems with synthetic molecules that evolution never prepared them to handle.

However, we have solutions. Using the bathtub metaphor again it's clear we want to reduce the outflow. But as with CO₂ the outflow is highly diluted and hard to capture. So we have to reduce the inflow.

There are roughly two ways to reduce the application of fertilizer. Together they can theoretically reduce our application by a factor of about ten. The first is using meat replacements instead of animals. To illustrate using phosphorus: in order to get 1 kg of phosphorus from our food we have to feed plants about 1.4 kg. If we feed plants to animals and eat the meat we need about 5.6 kg. (Yuan, 2021) Yes, you can use animal manure as fertilizer, but it is a pretty leaky bucket compared to not using animals at all. The second is precision application of fertilizer in just the right amount and at the right time, so less will be washed away instead of used by crops. Research shows that improvements in agricultural practices can reduce nitrogen losses from 52% to 22%, meaning less than half as much pollutes the environment. (You, 2023) This without compromising yields. Reduction to 10% losses seems possible before the soil is mined for nitrogen. (EUNEP, 2023) This type of precision farming is still considered too expensive but I think that will change.

Pesticides come in two categories. Insecticides that kill insects and herbicides that kill weeds. Instead of insecticides you can use nature by exposing the insects you don't want to their natural predators. The trick here is to use small lanes or plots with the right mix of crops. If you use really small plots it's called "pixel cropping" and the fields look completely different, both visually and in terms of biodiversity. Instead of herbicides you can use a weed killing robot. You can already buy them, although they are still expensive. (Upadhyay, 2024)

Personally I think that, apart from meat replacements, the answer will come from small autonomous agricultural robots. That may sound like a pie-in-the-sky fantasy, but allow me to quickly make the case. The potential impact is huge and once you look at it through a learning curves lens, it becomes very plausible. Computer intelligence and data communication are learning faster than anything else. It's only logical that in the coming decades little computers costing a few dollars will have more agricultural knowledge than a PhD. Paired with sensors, communication, and self-driving capabilities, I can't see how they would be unable to do the jobs described above. Weeding robots and their cousins are still expensive now, but costs are falling quickly. At the same time, the costs are rising for land, food, fertilizer, and the externalities we're finally recognizing. Falling and rising costs are bound to cross some time in the future. That conclusion, as a certain detective might say, is elementary.

Sidebar: The Chemical Load We're Only Beginning to Understand

In 2022, scientists at the Stockholm Resilience Centre published an update to the planetary boundaries framework. They added a new boundary: "novel entities." (Persson, 2022) This category includes synthetic chemicals, plastics, and other materials that don't occur naturally.

Their conclusion: we have already crossed this boundary. (Persson, 2022) The sheer quantity and variety of synthetic chemicals in the environment has reached levels that may cause irreversible harm to ecosystems and human health.

We produce roughly 350,000 synthetic chemicals. (Persson, 2022) Most have never been tested for environmental effects. (Persson, 2022) Many accumulate in food chains. Some disrupt hormones at very low concentrations. Microplastics now appear in human blood (Leslie, 2022), breast milk (Ragusa, 2022), and rain falling on remote mountaintops. (Brahney, 2020)

This is not an energy problem. Most of these chemicals come from agriculture (pesticides, herbicides), industry (manufacturing, mining), and consumer products (plastics, textiles). Electrifying the energy system helps only indirectly, by reducing the fossil fuel feedstocks used to make some of these chemicals.

If you want to reduce humanity's chemical footprint, solar panels are not where to look.

Why am I writing about food in a book about energy? Two reasons. First of all because I'm a system thinker. For me both food and fossil fuel are both energy sources. One for our

bodies. One for our machines. In both cases the problem is that getting them is unbalancing the natural world we depend upon. Separating them feels lazy and artificial. Secondly because critics of solar and wind sometimes worry about pollution from their manufacturing and waste. They point to the chemicals used in making semiconductors, the energy required to purify silicon, the waste products from battery production. These concerns are not baseless and I take my critics seriously. I know they mean well. But for that reason I want to help them to have a positive impact instead of causing harm.

The reality is that solar and wind don't exist in a vacuum. So you have to take the systems thinking approach and see how to reduce the damage of the system as a whole. You have to learn to *compare* interventions and see their impact on each others. With fossil fuels it's pretty clear cut. Since fossil fuels emit orders of magnitude more than solar and wind, slow-walking the expansion of solar and wind by asking "critical questions" leads us to use fossil fuels longer, and thus exacerbates the problems you are trying to reduce.

The impacts from food in terms of pollution are even worse than those of fossil fuel. If you really care about protecting the environment and your time and money are limited, you should focus on changing the food system, not squabbling over the impacts of solar panels. The nitrogen we spread on farm fields every year dwarfs the materials that go into solar panels. The pesticides we spray dwarf the chemicals in solar manufacturing. The habitat destruction from agriculture far exceeds any land use from solar farms.

This is not whataboutism. This is prioritizing. If you want to reduce humanity's pollution footprint, electrify energy and reform food. But of those two, food is the harder problem. We know how to generate clean electricity. We are still working out how to feed 8 billion people without drowning the world in fertilizer.

10.1.4 Comparing the Kilowatt-Hours

Not all energy sources are created equal. Every kilowatt-hour has a cost in human lives and environmental damage. The question is how big.

Researchers have calculated the deaths attributable to each energy source, including air pollution, accidents, and supply chain effects. (Ritchie, 2020) They have also calculated the lifecycle greenhouse gas emissions per unit of electricity generated. (Schlömer, 2014) The numbers are striking.

Sidebar: The Deadliest Energy Sources and Biggest Emitters

Deaths per gigawatt-hour of electricity generated: (Ritchie, 2020)

Source	Deaths per TWh
Coal	24.6
Oil	18.4
Biomass	4.6
Natural gas	2.8

Source	Deaths per TWh
Hydropower	1.3
Nuclear	0.05–0.5
Wind	0.04
Solar	0.02

A note on nuclear: The range reflects a large uncertainty. The lower estimate counts only direct deaths and uses conservative figures for long-term cancer mortality from accidents like Chernobyl. Higher estimates include best estimates for the excess deaths from radiation exposure over subsequent decades. What is not in dispute: nuclear's death toll is far lower than fossil fuels, even using the highest estimates.

Greenhouse gas emissions (grams CO₂-equivalent per kWh): ([Schlömer, 2014](#))

Source	g CO ₂ /kWh
Coal	740-1100
Natural gas	410-860
Biomass	430-660
Solar	41
Nuclear	12
Wind	11

A note on coal, natural gas and biomass: For coal the range reflects plant efficiency, and inclusion of life cycle analysis. For natural gas the range mostly depends on whether you use newer and higher aerial measurements for methane leakage or lower official estimates. For biomass indirect land use change is included.

Coal kills hundreds of times more people per unit of energy than solar, wind, or nuclear. It emits 20 to 75 times more greenhouse gases over the full lifecycle.

Natural gas is cleaner than coal but still far dirtier than renewables. Biomass, often marketed as "green," has significant emissions and health impacts from combustion.

Solar and wind occupy a different category entirely. Their health impacts come almost entirely from manufacturing and installation. Once operating, they produce nothing that enters anyone's lungs. Nothing that warms the climate. Nothing that runs off into rivers.

The comparison is not close. Every coal plant we shut down and replace with solar panels saves lives immediately. Not in 2050. Not when the climate stabilizes. Now, this year, in the lungs of people who would otherwise be breathing particulates.

When Chapter 6 argued that solar uses thousands of times less material than coal, it was talking about the inflow to our bathtub. When this chapter shows that solar produces hundreds of times fewer deaths, it is talking about the outflow.

With closed-loop recycling, the manufacturing emissions become a one-time cost, not a continuous burden. Build the panels. Run them clean for thirty years. Recycle the materials. Build new panels from the recovered silicon and aluminum. Each generation produces a smaller environmental footprint than the last.

10.1.5 We Already Proved We Can Do This

We have solved this kind of problem before. In 1985, scientists discovered a hole in the ozone layer over Antarctica. ([Farman, 1985](#)) The cause was traced to chlorofluorocarbons (CFCs), synthetic chemicals used in refrigeration and aerosols.

Two years later, 46 countries signed the Montreal Protocol, committing to phase out CFCs. ([Britannica, 2024](#)) Today, 197 countries have ratified it, making it the first treaty with universal participation. ([EPA, 2025](#))

Results:

- CFC concentrations in the atmosphere peaked in the late 1990s and have declined since ([UN, 2023](#))
- The Antarctic ozone hole stopped growing
- Full recovery is expected by approximately 2066 ([UN, 2023](#))

The Montreal Protocol prevented an estimated 2 million skin cancer cases per year by 2030. ([Dijk, 2013](#)) It also had climate benefits: CFCs are potent greenhouse gases. ([MIT, 2024](#))

This is not a story about environmental doom. It is a story about a problem identified, addressed, and solved. The same trajectory is possible for air pollution and climate change, if we choose it.

But in 1987, the world signed the Montreal Protocol, agreeing to phase out CFCs. It worked. CFC production dropped. The ozone layer stopped thinning. It is now slowly recovering. Scientists expect it to heal fully by the middle of this century.

The Montreal Protocol is the most successful environmental treaty in history. It demonstrates that humanity can identify a planetary boundary, agree to stop transgressing it, and actually follow through. We are capable of collective action on global environmental problems. We have done it before. We can do it again.

So we can limit raw mineral use until it's not a significant problem. We can get rid of the fossil fueled tyrants. And we can get rid of almost all pollution that kills countless people each year and causes climate change. But that brings us to our last and ultimate scarce resource: the surface of this beautiful planet. Will we have enough land?